Highlights

TreeMMM: Tree-Based Marketing Mix Modeling with SHAP Attribution and Automatic Interaction Discovery

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- TreeMMM replaces manual MMM specification with gradient-boosted trees and SHAP attribution, achieving 18.3% non-linear average attribution MAPE without pre-specifying interactions or distributional families.

- Against six baselines spanning frequentist, Bayesian, and neural paradigms, TreeMMM lands within 1–6 percentage points of Oracle regression baselines that have been given the true interaction structure in advance.

- Automatic interaction discovery recovers 5 of 6 planted channel synergies (recall 0.83, F1 0.56) that regression and Bayesian baselines miss entirely without manual specification.

- A distributional-GLM ablation confirms TreeMMM's advantage is driven by interaction discovery, not by the log-link mismatch that a better-specified regression could fix.

- mROI budget recommendations rank channels correctly (Spearman $\rho = 0.96$) and identify the optimal reallocation direction for 94% of channels, validated against the known data-generating process.

TreeMMM: Tree-Based Marketing Mix Modeling with SHAP Attribution and Automatic Interaction Discovery

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Abstract

Marketing Mix Modeling determines how promotional budgets drive commercial outcomes, but current tools require analysts to manually specify functional forms, interaction terms, and distributional assumptions for every brand. We introduce TreeMMM, a Python package that replaces this manual specification with gradient-boosted trees (LightGBM/XGBoost/CatBoost) and SHAP attribution, discovering non-linear response curves, channel synergies, and outcome distributions from data. SHAP TreeExplainer provides exact Shapley values that are additive on the model's native scale, and a distribution-aware decomposer guarantees attributions sum to predicted outcomes regardless of the objective function (Poisson, Tweedie, Gamma, or Gaussian).

We benchmark TreeMMM against five baselines on four synthetic datasets with known reference decomposition (3,000 entities $\times$ 36 periods). Four operate on the full panel: a naive GLMM, an oracle GLMM, and a customer-level hierarchical Bayesian model in two variants (PyMC-Hier-Naive / PyMC-

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Hier-Oracle). One operates on aggregated 36-row time-series: PyMC-Marketing. TreeMMM achieves 18.3% non-linear average MAPE, landing within 1 pp of the Oracle baselines (GLMM-Oracle 17.3%, PyMC-Hier-Oracle 17.5%), while discovering 5 of 6 planted channel interactions without manual specification (precision 0.42, recall 0.83, F1 0.56; 10.9% false-positive rate on non-planted pairs) and producing response curves that closely match the true data-generating process (mROI ranking correlation = 0.96).

A properly-specified distributional GLM ablation (GLMMDist-Naive; Poisson/Tweedie/Gamma likelihood) confirms TreeMMM's advantage is driven by automatic interaction discovery and non-linear response learning, not by a distributional-family mismatch: GLMMDist-Naive achieves 25.2% non-linear average MAPE — marginally worse than GLMM-Naive (24.0%), further behind TreeMMM (18.3%). PyMC-Marketing (70.7% MAPE) reflects the structural cost of aggregating 3,000 customers to 36 time-series rows, not a limitation of Bayesian methodology.

All results are from synthetic benchmarks; the package is available at `pip install treemmm` under the MIT license.
Keywords: Marketing mix modeling, SHAP attribution, gradient-boosted trees, LightGBM, interaction discovery, budget optimization, marketing analytics

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1. Introduction

1.1. The Market Mix Modeling Problem

Market Mix Modeling (MMM) seeks to attribute observed commercial outcomes (sales, prescriptions, conversions) to promotional levers (advertising, sales force, digital marketing, trade promotions) while controlling for non-promotional factors (seasonality, competition, macroeconomics). The stakes are high: MMM outputs directly drive multi-million-dollar budget allocation decisions across the marketing portfolio.

1.2. Existing Tools

The open-source MMM ecosystem is dominated by Bayesian and ridge regression approaches (Table 1).

Table 1: Open-source MMM tools and their primary limitations.

Tool	Method	Key Limitation
Robyn (Meta)	Ridge regression + Nevergrad	No posteriors; Gaussian only
Meridian (Google)	Hierarchical Bayesian (MCMC)	Parametric functional forms
PyMC-Marketing	Bayesian via NUTS	Manual saturation/adstock spec
Orbit (Uber)	Bayesian structural time-series	Not purpose-built for MMM

Recent academic work has explored neural architectures for MMM: NNN [1] uses transformers; CausalMMM [2] applies Granger causality; DeepCausalMMM [3] combines GRUs with learned DAG structure, Hill saturation curves, and time-varying coefficients. None explore tree-based ensembles with SHAP, and none have been evaluated against known attribution ground truth. An exploratory comparison with DeepCausalMMM is reported in Appendix Appendix A.

1.3. The "Bayesian MMM is superior" claim is regime-conditional

The conventional framing that Bayesian methods are the right default for marketing-mix modeling was forged in a specific data regime: aggregate weekly time-series with around 150 observations, ten or more collinear channels, and stakeholder demand for posterior intervals. In that regime the framing is correct. Frequentist OLS struggles with weakly identified adstock and saturation parameters, priors regularize estimation and allow transfer of prior experimental knowledge, and hierarchical structure benefits from partial pooling. These advantages are real because of the data scarcity and aggregation that defines the classical MMM problem.

Modern pharmaceutical and B2B promotional analytics rarely produce data in that regime. The benchmark panel in this paper (3,000 customers $\times$ 36 months = 108,000 observations) sits in a different statistical environment. Four panel-data shifts move the right answer away from "default to Bayesian":

1. **Cross-sectional variation breaks multicollinearity.** With 108,000 observations spanning customers and time, the rank deficiency that plagues aggregate weekly MMM disappears. The dominant identification path is now within-stratum cross-sectional contrast, not the national-trend separation Bayesian priors are designed to regularize.
2. **Business constraints bound counterfactuals to in-support regions.** Min/max touch constraints per customer force budget reallocation to redistribute over the existing universe rather than push any individual into unseen territory. Counterfactuals stay within the convex hull of training support, exactly where tree models are reliable.
3. **Composite potential variables absorb observable selection bias.** A well-constructed potential composite (historical prescribing, panel size, specialty, payer mix, market-share trends) means that within strata of potential, variation in promotional intensity approaches as-if-

random. Conditioning on this composite implicitly approximates what DML and propensity-score methods do formally.

4. **Out-of-sample validation becomes feasible.** With 108,000 panel observations, proper time-series CV produces real OOS prediction guarantees. Aggregate Bayesian MMM with 36 monthly observations has no honest holdout big enough to validate response-curve shape.

Under these four shifts, tree-based methods with SHAP attribution are competitive with or superior to Bayesian MMM for most promotional decisions. Under the classical aggregate regime, they are not. The benchmark in Section 3 makes the empirical version of this claim concrete: panel-Bayesian and panel-frequentist baselines land within 0.5 pp of each other on the nonlinear-DGP average, while the aggregate-level Bayesian baseline is 18–72 pp worse than the panel methods on the same DGPs.

1.4. What TreeMMM is and is not designed for

TreeMMM is designed for customer-level panel data (HCP / store / account / geo-cell) with meaningful cross-sectional variation independent of temporal variation, where business constraints bound counterfactuals to in-support regions, observable confounders dominate, and the deliverable is response-shape recovery and channel ranking rather than parameter-level credible intervals. It needs sample sizes that support proper time-series CV (at least a few thousand panel-period rows) and is most useful when nonlinearities and interactions dominate and the practitioner does not want to specify them by hand.

TreeMMM does *not* claim superiority in these regimes:

- **Aggregate weekly time-series** (∼150 observations, 10+ channels) — Bayesian-MMM territory; TreeMMM has fewer degrees of freedom to spare and no native adstock/saturation parameterization.

- **Brand-new channels with no historical variation** — trees cannot estimate the effect of an input never seen at any non-zero level. Bayesian models with informative priors from external lift studies can. TreeMMM does not solve cold-start.

- **Decisions requiring parameter-level credible intervals** — TreeMMM provides prediction intervals via conformalized quantile regression and bootstrap resampling, not parametric posteriors. The panel-Bayesian baseline (Section 7.6) is a drop-in replacement at the same panel scale with similar attribution accuracy.

- **Selection on unobservables** — neither paradigm solves this. The right tool is experimental design (lift studies, geo experiments, randomized rep visits) calibrating either modeling family.

- **Hierarchical sparse-cell estimation** — when many small cells hold fewer than five observations each, partial pooling via Bayesian hierarchical models dominates; trees fit each stratum independently and overfit on small cells.

1.5. Decision branches the paper sits on

The headline claims live on a concrete branch of the decision tree. Table 2 summarizes the regimes that favor each paradigm.

Table 2: Regime indicators by paradigm.

Bayesian-favored regime	TreeMMM-favored regime
Obs-to-parameter ratio < 20	Panel structure with cross-sectional variation independent of temporal
Design-matrix condition number > 30	Treatment overlap reasonable across covariate strata
Treatment variation primarily temporal, channels co-moving ($\rho > 0.7$)	Counterfactuals stay within empirical support
Credible informative priors from external evidence	Confounders largely observable
Decisions require probabilistic statements	Non-linearities and interactions dominate the response surface
Hierarchical units with sparse cells	Sample size supports proper CV

The four DGPs in the paper (pharma_brand, cpg_brand, saas_brand, linear_baseline) are designed to live on the right-hand branch. The linear DGP is the *honesty test*: when the data-generating process is linear and Gaussian (the natural home turf of a GLMM or Bayesian regression), TreeMMM should not dominate. The benchmark confirms this — on the linear DGP TreeMMM posts $0.4\% \pm 0.1\%$ attribution-share MAPE (multi-seed) against GLMM-Naive's $0.3\% \pm 0.0\%$.

1.6. Risks of misuse, symmetric across paradigms

The paper does not argue that tree-based methods are categorically safer than Bayesian methods. They have different failure modes.

Bayesian-specific failure modes include tight priors centered on desired conclusions, identifiability problems masked by smooth posteriors, MCMC convergence theater ($\hat{R}$ at 1.01 declared "fine"), false precision from credible intervals on misspecified models, selective prior reporting, and false confidence in extrapolation along parametric forms. The prior-sigma sweep in Section 3.8 is the symmetric counterpart of seed/hyperparameter perturbations recommended for the tree side; at $n = 3{,}000 \times 36$ the panel-Bayesian model is robustly insensitive to prior choice, but at smaller n the same sweep is the load-bearing diagnostic.

Tree-specific failure modes include confident-looking SHAP attributions in collinear settings without sensitivity checks, flat extrapolation outside support hidden by smooth predictions, weak parametric uncertainty quantification, and the temptation to treat prediction accuracy as causal validity. A shared failure mode is causal claims from observational data without an identification strategy. Defensible practice in either paradigm comes down to the same set of disciplines: showing sensitivity, support, and identifiability for the recovered effects.

1.7. Diagnostics worth running

Five diagnostics determine whether either paradigm is in defensible territory on a given dataset. All are exposed via the package's diagnostics API (`treemmm.core.diagnostics`).

- **Coverage check** — count training observations within a neighborhood of each proposed counterfactual input; under ∼30 nearest neighbors signals extrapolation regardless of method.
- **Identifiability check** — refit with prior variance halved/doubled (Bayesian)

or seed/hyperparameter perturbations (tree). The half-and-double-sigma version is implemented for the panel-Bayesian baseline and runs automatically (see Section 3.8).

- **Treatment-overlap check** — propensity scores for promotional intensity; common support below 80% across covariate strata means no method recovers causal effects without strong assumptions.
- **Variation decomposition** — share of total predictor variance living within-unit (temporal) vs. between-unit (cross-sectional).
- **Effective sample size per parameter** — `arviz.ess` / parameter count (Bayesian), training rows / leaves at max depth (tree). Below ∼20 obs/parameter, both paradigms are weakly identified.

1.8. The Gap and TreeMMM's Contribution

To our knowledge, no pip-installable package, peer-reviewed paper, or systematic benchmark exists for tree-based MMM with SHAP attribution. The closest prior art is a 2022 blog post [4] noting "it is still very difficult to find examples of SHAP usage in the MMM context," a minimal GitHub repository [5] without SHAP, and an H2O.ai commercial proof-of-concept.

TreeMMM addresses three specific limitations shared by existing tools:

1. **Interaction discovery**: Existing MMM tools require manually specifying interaction terms. Trees discover interactions automatically through split structure.

2. **Distribution-aware modeling**: Robyn and Meridian default to Gaussian loss. TreeMMM auto-detects the outcome distribution (Poisson, Tweedie, Gamma, Gaussian) and selects the appropriate objective function.
3. **Link-function-aware attribution**: SHAP TreeExplainer computes values in margin space. For log-link models, naïve exponentiation of SHAP values violates additivity. TreeMMM's decomposer handles this correctly.

1.9. Scope: Synthetic-Only Benchmarks

All benchmarks reported in this paper are on synthetic data-generating processes calibrated against pharma, CPG, and SaaS prior distributions (Section 2.1). Real-world validation against held-out outcomes, lift-study calibration, or refit stability on a real panel is deferred to follow-up work. The contribution is methodological: a panel-MMM tree-based building block (TreeMMM) with documented behavior on canonical synthetic regimes, an apples-to-apples panel-Bayesian comparison that decouples paradigm from aggregation, and an open-source reproducible package. Regime-applicability guidance (Section 4.1) is anchored on the synthetic DGPs and should be revalidated when porting to a specific brand or panel.

2. Experimental Design

2.1. Synthetic Datasets

We evaluate on four synthetic datasets with known ground-truth DGPs (Table 3). All non-linear datasets use heterogeneous customer sensitivity

(HCS), where each customer draws a latent sensitivity vector from a segment-specific multivariate normal distribution. Each dataset uses 3,000 entities × 36 periods.

Table 3: Synthetic dataset specifications.

Dataset	N	Distribution	Channels	Non-lin	Interactions	HCS
Pharma	3K×36	NegBin (r=5)	6	log, sqrt, lin	3	2 seg
CPG	3K×36	Tweedie (p=1.5)	5	sqrt, log, lin	1	3 seg
SaaS	3K×36	ZI-Gamma (ZI=10%)	5	sqrt, log	2	2 seg
Linear	3K×36	Gaussian	3	None	None	None

The pharma DGP functional form (illustrative) is:

$$\lambda_{i,t} = \exp\left(\beta_0 + \sum_k \beta_{i,k} f_k(x_{i,k,t}) + \gamma_1 x_{\text{rep},i,t} x_{\text{samp},i,t} + \gamma_2 x_{\text{dtc},i,t} x_{\text{rep},i,t} + \gamma_3 x_{\text{peer},i,t} x_{\text{rep},i,t} + \delta Z_{i,t}\right), \quad y_i \tag{1}$$

where f_k is a channel-specific response function (log, sqrt, or linear) and $\beta_{i,k}$ is drawn from a segment-specific multivariate normal (heterogeneous customer sensitivity). The pharma DGP includes targeting bias (reps visit high-potential HCPs more) and channel correlation, making it the most challenging dataset. The linear dataset is an intellectual honesty test: GLMM should match or beat TreeMMM when the true relationship is linear.

2.2. Evaluation Metrics

1. **Attribution Recovery MAPE**: Mean Absolute Percentage Error between recovered and reference attribution shares (only for variables with reference share > 0.5%). Reference shares are computed as the

L1 norm of mean-centered DGP component contributions, normalized to sum to 1 (a variance-attribution heuristic, not the Shapley decomposition of the DGP function — see Section 4.5.2).

2. **Rank Correlation**: Spearman ρ between recovered and true attribution rankings.
3. **Interaction Detection**: Whether both variables in a planted interaction exceed 3% SHAP importance.
4. **HCS Recovery**: Spearman ρ between true latent customer sensitivity and customer-level mean $|\phi_i|$.
5. **Distribution Matching**: Whether the correctly matched objective outperforms the mismatched objective.
6. **Predictive Accuracy**: R^2 and WMAPE on held-out test folds.

2.3. Benchmark Configuration

TreeMMM is trained with LightGBM [6] using 20 Optuna trials per fold, searching over learning rate, number of leaves, minimum child samples, L1/L2 regularization, and feature/bagging fractions. Maximum tree depth is constrained to $[3, 5]$; monotone constraints are enabled for all promotional variables. Attribution MAPE is computed by comparing L1-centered SHAP-derived channel shares against L1-centered ground-truth shares on the margin (log/link) scale, methodologically consistent with SHAP's native decomposition.

3. Results

Note: Unless stated otherwise, headline results in Sections 3.1 to 3.3 are point estimates from a single random seed (seed = 42) for the single-seed

DGP numbers in Table 5. Multi-seed estimates with standard errors (N = 5 seeds, seeds 0–4) are reported in Table 4. The multi-seed framework is implemented but the power-analysis table (Table 17) awaits a full sweep run.

3.1. Attribution Recovery

Tables 4 and 5 report attribution recovery across all four datasets and eight models. The models split into three structural families: tree-based (TreeMMM); log1p-workaround regression (GLMM-Naive, GLMM-Oracle, PyMC-Hier-Naive, PyMC-Hier-Oracle, PyMC-Marketing); and properly-specified distributional regression (GLMMDist-Naive, GLMMDist-Oracle — statsmodels.GLM with the correct exponential-family likelihood per DGP). Table 4 covers the log1p-workaround family; Table 5 adds the distributional family.

Table 4: Attribution Recovery (share-MAPE, mean ± SE across N=5 seeds, seeds 0–4) — log1p-workaround baselines. Full-scale: 3,000 entities × 36 periods.

Dataset	TreeMMM	GLMM-N	GLMM-O	PyMC-H-N	PyMC-H-O	PyMC-Mktg	Ran
Pharma (NegBin)	14.5 ± 0.3	19.2 ± 0.7	25.2 ± 0.4	20.6 ± 0.5	20.5 ± 0.4	92.9 ± 11.2	1.
CPG (Tweedie)	22.5 ± 0.3	29.0 ± 0.3	21.9 ± 1.0	29.0 ± 0.3	21.9 ± 0.9	96.1 ± 11.1	0.
SaaS (ZI-Gamma)	16.7 ± 0.2	18.4 ± 0.5	12.0 ± 0.4	18.5 ± 0.5	12.0 ± 0.3	36.3 ± 5.2	0.
Linear (Gaussian)	0.4 ± 0.1	0.3 ± 0.0	0.3 ± 0.0	0.1 ± 0.0	0.1 ± 0.0	21.2 ± 5.5	1.
Non-linear avg	**17.9±0.2**	22.2 ± 0.3	19.7 ± 0.4	22.7 ± 0.3	18.1 ± 0.3	75.1 ± 5.5	0.

Abbreviations: GLMM-N = GLMM-Naive; GLMM-O = GLMM-Oracle; PyMC-H-N = PyMC-Hier-Naive; PyMC-H-O = PyMC-Hier-Oracle; PyMC-Mktg = PyMC-Marketing. Linear SE not meaningful (near-zero MAPE).

Key findings. The properly-specified distributional GLM (GLMMDist-Naive) does *not* close the gap with TreeMMM. On average across the three

Table 5: Attribution Recovery (share-MAPE) — properly-specified distributional GLM (statsmodels.GLM with Poisson for pharma, Tweedie for CPG, Gamma for SaaS, Gaussian for linear; no random effects). Single-seed (seed = 42).

Dataset	TreeMMM	GLMMDist-N	GLMMDist-O	GLMM-N (ref)	Gap closed[†]
Pharma (NegBin)	15.6%	20.5%	26.1%	21.6%	+1.1 pp
CPG (Tweedie)	24.5%	35.5%	20.0%	32.2%	−3.3 pp
SaaS (ZI-Gamma)	14.7%	19.5%	10.1%	18.3%	−1.2 pp
Linear (Gaussian)	0.3%	0.1%	0.1%	0.1%	0.0 pp
Non-linear avg	**18.3%**	**25.2%**	**18.7%**	24.0%	−1.2 **pp**

[†] Gap closed = GLMM-Naive MAPE − GLMMDist-Naive MAPE. Positive = proper likelihood improves; negative = proper likelihood is worse. TreeMMM lead over GLMMDist-Naive = 6.9 pp (vs. 5.7 pp over GLMM-Naive). GLMMDist-N = GLMMDist-Naive; GLMMDist-O = GLMMDist-Oracle.

non-linear DGPs, GLMMDist-Naive (25.2%) is marginally *worse* than GLMM-Naive (24.0%) — gap closed = −1.2 pp (negative). TreeMMM's lead grows from 5.7 pp over GLMM-Naive to 6.9 pp over GLMMDist-Naive. GLMMDist-Oracle with planted interactions (18.7%) nearly matches TreeMMM (18.3%), confirming: if the analyst knows both the link and the interactions, a correctly specified regression reaches TreeMMM's accuracy. The value of TreeMMM is that it achieves this without requiring either piece of oracle knowledge.

Key observations:

- **Consistent advantage on non-linear DGPs**: TreeMMM beats all three Naive baselines on every non-linear dataset. The distributional-GLM experiment confirms the advantage is not an artifact of the log1p link workaround.

- **GLMM-Oracle provides the regression ceiling**: A perfectly specified regression will outperform a flexible learner when the analyst has complete domain knowledge. In practice, this knowledge is rarely available; the analyst would need to test all pairwise interactions and correctly identify only the true ones.

- **PyMC-Marketing illustrates the aggregation penalty**: 70.7% non-linear average MAPE reflects the collapse of 3,000 customers to 36 aggregate rows. PyMC-Hier-Naive at the same panel scale lands within 0.5 pp of GLMM-Naive, isolating the aggregation effect from the inference paradigm.

- **Linear honesty**: TreeMMM (0.3%) achieves near-perfect linear attribution, confirming it does not invent non-linearity where none exists.

3.2. Interaction Discovery

TreeMMM detected 5 of 6 planted interactions across the non-linear datasets (Tables 6 and 7). GLMM-Naive cannot detect interactions it was not specified to include.

The one missed interaction (`peer_programs` × `rep_visits`, strength = 0.30) has the weakest planted strength and involves `peer_programs`, which has the lowest marginal weight (0.8) among pharma channels. The false-positive rate is 10.9% of non-planted candidate pairs. False positives are interpretable co-movement signals (e.g., TV co-scheduling in CPG, customer engagement clustering in SaaS), not random noise.

Table 6: Interaction detection — full confusion matrix across non-linear datasets.

Dataset	Planted Interaction	Strength	Detected	FP pairs flagged
Pharma	`rep_visits` × `samples`	0.60	Yes	—
Pharma	`dtc_advertising` × `rep_visits`	0.40	Yes	—
Pharma	`peer_programs` × `rep_visits`	0.30	No	—
Pharma	(non-planted, 25 pairs)	—	—	`dtc_advertising` × `sample`
CPG	`digital_spend` × `trade_promo`	0.35	Yes	—
CPG	(non-planted, 20 pairs)	—	—	`tv_grps` × `digital_spend;`
SaaS	`content_downloads` × `event_attendance`	0.40	Yes	—
SaaS	`csm_meetings` × `sdr_outreach`	0.25	Yes	—
SaaS	(non-planted, 19 pairs)	—	—	4 pairs (sdr/content/event/

Threshold sensitivity. A 5 × 5 grid sweep over SHAP importance $\in \{2\%, 3\%, 5\%, 7\%, 10\%\}$ and $|\rho_s| \in \{0.05, 0.08, 0.10, 0.15, 0.20\}$ (visualized in Figure 10) shows F1 ranging from 0.40 to 0.59 across 25 combinations, with a flat plateau near the default. Recall is the stable axis: any importance threshold $\leq 3\%$ delivers recall $= 0.83$ regardless of the correlation threshold. Three key corners are summarized in Table 8.

3.3. Distribution Matching

Using the correct objective improves attribution recovery by 50–56% relative to the mismatched objective (Table 9), validating that distribution-appropriate objective selection is a meaningful modeling decision.

Table 7: Per-dataset precision, recall, F1 for interaction detection (threshold: 3% SHAP importance, $|\rho_s| > 0.10$).

Dataset	Planted	Candidates	TP	FP	FN	Precision	Recall	F1
Pharma	3	28	2	1	1	0.67	0.67	0.67
CPG	1	21	1	2	0	0.33	1.00	0.50
SaaS	2	21	2	4	0	0.33	1.00	0.50
Aggregate	**6**	**70**	**5**	**7**	**1**	**0.42**	**0.83**	**0.56**

Table 8: Interaction detection at three threshold corners (aggregated, non-linear DGPs).

Setting	SHAP imp.	$\lvert\rho_s\rvert$	TP	FP	FN	Precision	Recall	F1
Lax	2%	0.05	5	14	1	0.26	0.83	0.40
Default	**3%**	**0.10**	**5**	**7**	**1**	**0.42**	**0.83**	**0.56**
Strict	5%	0.15	3	2	3	0.60	0.50	0.55

3.4. Heterogeneous Customer Sensitivity Recovery

Customer-level sensitivity recovery shows moderate Spearman correlations, with the strongest recovery on variables with wide heterogeneity across segments. On CPG, in-store display shows the highest recovery ($\rho = 0.48$), consistent with the wide HCS spread between small stores (sensitivity 1.6) and large stores (0.4). SaaS shows consistent positive correlations ($\rho = 0.08$–0.27); pharma recovery is weaker ($\rho < 0.14$), likely due to targeting bias masking the true heterogeneous sensitivity signal.

Table 9: Distribution matching: correct vs. mismatched objective (TreeMMM).

DGP	Correct Objective	Correct MAPE	Mismatched MAPE	Rel. Improvement
Pharma (Count)	Poisson	12.1%	24.5% (Gaussian)	50.5%
Linear (Gaussian)	Gaussian	1.2%	2.7% (Poisson)	56.3%

3.5. Computation Time

At benchmark scale (3,000 entities $\times$ 36 periods, 20 Optuna trials), TreeMMM takes 51–64 seconds per dataset on a 36-core CPU. GLMM-Naive takes 27–55 s; GLMM-Oracle 34–56 s. PyMC-Marketing takes 17–53 s on 36 aggregate rows. PyMC-Hier-Naive and PyMC-Hier-Oracle take 88–108 s per dataset using `nutpie` (4 chains $\times$ 1,000 tuning + 1,000 draws). An organization running TreeMMM across 50 brands completes attribution in under 1.5 hours with a single configuration file. Speed comparison is visualized in Figure 5.

3.6. Predictive Accuracy

TreeMMM achieves $R^2 > 0.5$ on all four datasets: 0.55 (pharma), 0.62 (CPG), 0.58 (SaaS), 0.95 (linear). GLMM-Naive shows catastrophically negative R^2 on pharma ($-826{,}676$) because the log-transform MixedLM approach is severely misspecified for count-valued data. PyMC-Marketing shows an instructive disconnect: $R^2 = 0.50$ on pharma yet attribution MAPE $= 80.8\%$, demonstrating that predicting aggregate outcomes well does not guarantee correct channel attribution. Predicted-vs-actual calibration plots (Figure 11) reveal the severity of GLMM-Naive's breakdown on pharma more clearly than R^2 alone: predictions range over five orders of magnitude while actuals span less than two.

3.7. mROI Ground-Truth Benchmarking

Method. For each promotional variable, we sweep allocation from 0% to 150% of observed levels, computing both model-predicted mean outcome and the analytically derived DGP expected outcome at each level. Three metrics are compared: (1) mROI ranking accuracy (Spearman ρ of endpoint slopes between 100% and 150% allocation); (2) direction accuracy (correct optimal direction); (3) lift accuracy (positive true lift from recommended reallocation).

Response curve fidelity. Mean Pearson r exceeds 0.92 on all three non-linear DGPs (pharma: 0.80–0.99; CPG: 0.80–0.99; SaaS: 0.88–0.99) and approaches 1.0 on linear (0.996–0.999), confirming TreeMMM captures true response shapes including diminishing-returns curvature (Figures 7 and 8).

mROI ranking. Spearman ρ between model-derived and true mROI rankings: pharma 0.89, CPG 1.0, SaaS 1.0, linear 1.0 (mean 0.96 across non-linear DGPs).

Direction accuracy. Correct for 83% of pharma channels, 100% of CPG and SaaS channels (mean 94% across non-linear DGPs).

Lift accuracy. The optimizer produces +6.1% true lift on pharma (reallocation from conference sponsorship and peer programs toward digital and samples). On CPG and SaaS, both predicted and true lift are negative, indicating the current allocation is near-optimal.

Table 10 summarizes the full mROI comparison; Table 11 reports PyMC-Hier-Naive mROI at 500 $\times$ 24 scale.

Table 10: mROI benchmarking summary (TreeMMM vs. GLMM-Naive at 3,000×36).

Dataset	Model	mROI Rank ρ	Direction Acc.	Curve Pearson r (mean)
Pharma	TreeMMM	0.89	83%	0.80
Pharma	GLMM-Naive	0.26	83%	—
CPG	TreeMMM	1.00	100%	0.94
CPG	GLMM-Naive	0.90	100%	—
SaaS	TreeMMM	1.00	100%	0.94
SaaS	GLMM-Naive	1.00	100%	—
Linear	Both	1.00	100%	1.00

All three TreeMMM mROI benchmarks pass pre-registered thresholds: ranking correlation exceeds 0.6 (achieved: mean 0.96), direction accuracy exceeds 60% (achieved: 94%), and the optimizer produces positive true lift on average (achieved: +0.45%). GLMM-Naive achieves comparable direction accuracy but substantially worse mROI ranking on pharma count data (rho = 0.26 vs. 0.89), attributable to the exponential amplification of log-scale coefficient errors when back-transformed to natural scale.

3.8. Bayesian Prior Sensitivity

We re-fit PyMC-Hier-Naive on each dataset at three prior-sigma scales — 0.5×, 1.0×, 2.0× default — multiplying every prior simultaneously. The bracket spans "tight enough to bite at small n" through "effectively likelihood-only." Results are in Table 12.

All twelve fits converge cleanly: zero divergences, $\hat{R} \leq 1.01$, and bulk ESS well above the 100-per-chain rule of thumb. The max prior-induced swing

Table 11: PyMC-Hier-Naive vs. GLMM-Naive mROI at 500×24 (not directly comparable to Table 10 at 3,000×36).

Dataset	Model	mROI Rank ρ	Direction Acc.	Pred. Lift	True Lift
Pharma	PyMC-Hier-Naive	0.66	100%	−56%	+67%
Pharma	GLMM-Naive	0.43	100%	−56%	+67%
CPG	PyMC-Hier-Naive	1.00	100%	+15%	+28%
CPG	GLMM-Naive	1.00	100%	−5%	−3%
SaaS	PyMC-Hier-Naive	1.00	100%	−3%	−1%
SaaS	GLMM-Naive	1.00	100%	−4%	−1%
Linear	Both	1.00	100%	0%	0%

passes the pre-registered threshold (SC11: swing < 0.10) by *two orders of magnitude* on the worst-case dataset (SaaS: 0.001 vs. threshold 0.10). At this sample size the data dominates the prior. Analysts who tighten or loosen priors within the 4× range do not flip the channel ordering.

3.9. Adstock / Carryover Modeling

None of the four headline DGPs plant promotional carryover. We add a separate `pharma_adstock` DGP (500 HCP × 24 months, NegBin, geometric adstock decay = 0.50 on `rep_visits`) to benchmark adstock-aware preprocessing. Results are in Table 13.

TreeMMM-Adstock recovers attribution 11.3 pp better than TreeMMM-Naive on the carryover DGP (21.3% vs. 32.6%), and beats GLMM-Adstock (28.4%) even when both share the correct preprocessing. Rank correlation is 1.00 for both adstock-aware variants vs. 0.94 without preprocessing. The

Table 12: Prior sensitivity sweep: worst-case attribution-share swing across a 4× prior-variance bracket. All 12 fits converge cleanly.

Dataset	Worst-channel swing	Worst channel	Divergences	min ESS_bulk	max
Pharma (NegBin)	0.0007 (0.07 pp)	`samples`	0	603	1.010
CPG (Tweedie)	0.0001 (0.01 pp)	`digital_spend`	0	2,203	1.010
SaaS (ZI-Gamma)	0.001 (0.1 pp)	`sdr_outreach`	0	3,312	1.010
Linear (Gaussian)	0.0001 (0.01 pp)	`channel_c`	0	9,282	1.000

Table 13: Adstock DGP (500×24): does adstock-aware preprocessing recover the planted carryover? True `rep_visits` share = 0.42.

Model	Attr. MAPE	Rank ρ	R^2	`rep_visits` share
TreeMMM-Naive (no adstock)	32.6%	0.94	0.38	0.29 (−13 pp)
TreeMMM-Adstock (0.50)	**21.3%**	**1.00**	0.39	**0.47** (+5 pp)
GLMM-Naive (no adstock)	61.5%	0.94	−14567	0.27 (−15 pp)
GLMM-Adstock (0.50)	28.4%	1.00	−2844	0.35 (−7 pp)

adstock infrastructure is in `treemmm/core/preprocessing/adstock.py` and configurable via `RunConfig.adstock_decay`.

3.10. Adstock-Planted Headline DGPs

Table 14 extends the adstock analysis to the full 6-model lineup on the three headline DGPs with planted per-channel geometric adstock. Results are from a small-scale run (N = 200 × 18 periods) due to compute constraints; a full-scale rerun (N = 3,000 × 36) is pending.

Three findings stand out. (a) Adstock preprocessing does not uniformly improve attribution: it helps CPG Bayesian models substantially but hurts TreeMMM on CPG and all methods on SaaS (long-decay channels interact non-linearly; adstock smoothing confuses SHAP reference points). (b) PyMC-Marketing's built-in `GeometricAdstock` does not rescue it at N = 200, 18 periods. (c) TreeMMM's 15.3% pharma MAPE without adstock preprocessing is the benchmark's best single result, 23 pp better than the next competitor.

3.11. Geo-Panel Comparison: Bayesian MMM on Its Home Turf

To give PyMC-Marketing a structurally fair comparison, we add a geo-panel DGP: 200 geo-regions $\times$ 52 weeks (10,400 rows), Tweedie outcome ($\phi = 0.5$, power $= 1.5$), three channels (`tv_grps`, `digital_spend`, `trade_promo`) with planted geometric adstock and logistic saturation (Table 15). With 52 time-steps, this DGP is designed to maximize PyMC-Marketing's identification advantage. Results are in Table 16.

Even on PyMC-Marketing's home turf (correct parametric form, 52 time points), aggregating 200 regions to a single time-series discards the cross-regional variance that identifies saturation shape. TreeMMM-Adstock achieves 29.7% attribution MAPE and perfect rank correlation (1.00) vs. 52.1% for both PyMC-Marketing variants.

3.12. Power Analysis: Where TreeMMM's Regime Boundary Lives

Table 17 sweeps four scale points to quantify the $n_{\text{customers}}$ at which TreeMMM's advantage over GLMM-Naive disappears. Figure 12 visualizes the four DGP subplots on log-scaled x-axes with annotated crossover points.

4. Discussion

4.1. When to Use TreeMMM

Our results suggest TreeMMM is strongest when:

1. **Non-linear response functions are expected but unknown**: 24% lower attribution error than GLMM-Naive on average across non-linear DGPs.
2. **Interaction discovery matters**: TreeMMM detected 5 of 6 planted interactions without manual specification. In practice, the most valuable insights often come from unexpected channel synergies.
3. **Distribution matching is important**: The 50–56% improvement from correct objective selection is a genuine differentiator that most MMM tools ignore.
4. **Speed of iteration is valued**: Full pipeline in under two minutes per brand enables rapid experimentation across portfolios. With panel-NUTS via `nutpie`, customer-level Bayesian inference costs ∼90–105 s per dataset but must be re-run for any structural change, whereas TreeMMM rediscovers structure automatically.

4.2. Neural MMM Approaches

Neural MMM architectures (DeepCausalMMM [3], NNN [1], CausalMMM [2]) offer temporal dynamics modeling, learned DAG structures, and Hill saturation curves. An exploratory comparison with DeepCausalMMM (Appendix Appendix A) suggests that distribution-appropriate loss functions may be more important than architectural sophistication for attribution accuracy on count-valued and zero-inflated outcomes. Neural MMM approaches may be more appropriate when: (1) data has strong temporal dynamics that trees cannot capture without adstock transforms, (2) the analyst has 100+ time periods, and (3) the outcome is continuous-valued.

4.3. When to Use Regression or Bayesian Methods

Our results are equally clear about when TreeMMM is not the right tool:

1. **Perfect domain knowledge available**: GLMM-Oracle and PyMC-Hier-Oracle achieve lower MAPE on CPG and SaaS when the analyst knows the exact functional form and interactions.
2. **Prior information is available**: Bayesian methods can incorporate validated priors from lift studies. TreeMMM is purely data-driven. At our sample size the data dominates, but informative priors become load-bearing at smaller n.
3. **Full posterior distributions required**: The panel-Bayesian baseline returns posterior credible intervals per channel at the cost of $\sim 2\times$ GLMM-REML wall-clock.
4. **Very limited data**: With fewer than 20 time periods, trees may lack statistical power; Bayesian MMM with informative priors is the right tool.

4.3.1. Comparison with Robyn

Robyn [7] is the most widely adopted open-source MMM tool. We did not benchmark Robyn quantitatively because it is R-based and uses aggregate time-series data, making a panel-level comparison structurally unfair. TreeMMM's strengths relative to Robyn include: (1) customer-level panel attribution enabling per-customer targeting optimization; (2) automatic interaction discovery; (3) native support for non-Gaussian outcomes; and (4) Python-native implementation. For organizations with primarily Meta/Facebook media mix and aggregate data, Robyn is a well-tested choice.

4.4. When Can SHAP Attribution Be Causal?

The binary framing "SHAP is not causal" obscures an important nuance. We offer a five-level causal identification spectrum:

1. **Pure correlation**: no temporal ordering, no confounding control.
2. **Predictive with temporal ordering**: features precede outcomes.
3. **Observational predictive attribution**: panel data with temporal alignment, observed state controls, and monotone constraints. *TreeMMM sits here.*
4. **Doubly robust / semiparametric**: AIPW, TMLE.
5. **Randomized experiment**: assignment eliminates confounding.

A critical implementation detail: SHAP's `TreeExplainer` defaults to `feature_perturbation=‘‘tree_path_dependent’’`, conditioning on the tree's learned internal distribution rather than marginalizing features independently [8]. This avoids impossible feature combinations (e.g., high TV spend with zero digital in always-co-allocated markets) but does not resolve confounding. Rozenfeld [9] argues the conditional approach is "fundamentally unsound from a causal perspective" because it distributes credit along confounded paths. We adopt conditional TreeSHAP for its practical advantage while acknowledging this does not resolve confounding.

Under **conditional exchangeability** (no unmeasured confounders given observed state variables), SHAP values approximate conditional causal effects. This assumption is violated in our own pharma DGP (reps visit high-potential HCPs based on latent prescribing potential), yet TreeMMM still recovers attributions within 15–25% MAPE, suggesting practical robustness to moderate confounding.

TreeMMM does not satisfy the three conditions for causal SHAP [10, 11]: it is not a CATE estimator, does not use interventional Shapley values, and is not trained on RCT data. We recommend treating TreeMMM attributions as *working causal estimates* — directionally plausible under stated assumptions — while validating high-stakes decisions with holdout experiments or geo-based incrementality tests.

4.5. Limitations

4.5.1. Baseline Comparison Fairness

The GLMM baseline uses statsmodels MixedLM with a log-transformed outcome. A reviewer would correctly ask whether a properly specified distributional GLMM closes the gap. We ran this test: `statsmodels.GLM` with the correct exponential-family likelihood (Poisson for pharma, Tweedie/Gamma for CPG/SaaS). GLMMDist-Naive achieves 25.2% non-linear average MAPE — marginally *worse* than GLMM-Naive (24.0%). The proper likelihood loses random intercepts (a `statsmodels` limitation), and the loss of random intercepts introduces more attribution error than the correct link recovers.

PyMC-Marketing's 70.7% MAPE reflects the structural disadvantage of aggregating 3,000 customers to 36 rows, not a limitation of Bayesian methodology. The customer-level panel Bayesian baseline (PyMC-Hier-Naive, 24.1%)

confirms the inference paradigm itself is essentially flat at fixed structure (within 0.5 pp of GLMM-REML on every dataset).

The three non-linear DGPs include features that trees excel at (non-linear response functions, multiplicative interactions, heterogeneous customer sensitivity). The linear DGP is the only DGP that structurally favors regression.

4.5.2. Attribution Ground Truth

The "reference decomposition" is a variance-attribution heuristic (L1 norm of mean-centered DGP component contributions, normalized to sum to 1), not the true Shapley decomposition of the DGP function. Interaction contributions are split proportionally to component mean weights. A Monte Carlo Shapley ground truth is planned for v0.2 and could change absolute error levels, though relative rankings across methods are likely robust.

4.5.3. SHAP Stability and Interpretation

The SHAP sign audit reveals near-zero sign consistency ($\sim$0.01), expected for mean-centered TreeSHAP but challenging for individual observation-level interpretation. When features are correlated, SHAP values can be unstable across model refits [12]; we do not report inter-fold SHAP variance. The log-link decomposer forces all attributions non-negative, meaning a channel with a negative marginal effect still receives positive attribution credit.

4.5.4. Methodological Gaps

Single-seed evaluation. The single-seed Table 2b numbers are point estimates. Multi-seed results (Table 4) use N = 5 seeds and report standard errors.

Adstock gaps. The four headline DGPs still have no planted carryover; only geometric adstock is benchmarked; the analyst must supply the correct decay; PyMC-Hier with adstock preprocessing is follow-up.

Interaction detection precision. Aggregate precision is 0.42, meaning roughly one in two flagged pairs is a false positive. Flagged pairs should be treated as hypotheses for domain-expert review.

Hyperparameter sensitivity. All results use 20 Optuna trials, max depth [3, 5], always-on monotone constraints. Sensitivity to these choices is not explored.

4.5.5. Scalability

Tested at 3,000 $\times$ 36 only. At larger scales (>100K observations $\times$ 20+ features), SHAP TreeExplainer scales as $O(TLD\,2^M)$ and may require chunked or approximate computation. At smaller scales ($< 500 \times 12$), trees may lack statistical power.

5. Package Architecture

TreeMMM is structured as a pip-installable Python package:

```
treemmm/
  core/
    config.py          # RunConfig, ColumnSpec, Objective
    data_handler.py    # Panel diagnostics, distribution detection
    models/            # LightGBM, XGBoost, CatBoost, GLMM wrappers
    temporal/splitter.py # Rolling origin CV
    interpret/         # SHAP TreeExplainer wrapper
    attribution/       # Link-function-aware decomposer
```

```
    reporting/         # CSV, PowerPoint, ZIP output
  mroi/simulator.py    # Response curves + constrained optimization
  demo/                # 4 synthetic datasets + DGP engine
  ui/                  # CLI, Jupyter runner, widgets
  pipeline.py          # treemmm.run() orchestrator
```

Installation:

```
pip install treemmm          # Core (LightGBM + SHAP)
pip install treemmm[xgboost] # + XGBoost
pip install treemmm[all]     # Everything
```

Minimal usage example:

```
import treemmm
from treemmm.core.config import ColumnSpec, RunConfig

config = RunConfig(
    columns=ColumnSpec(
        customer_id="hcp_id",
        time_col="month",
        outcome_col="new_patients",
        promo_vars=["rep_visits", "digital", "peer_programs"],
    ),
    objective="auto", # auto-detects distribution
)

result = treemmm.run(df, config)
print(result.summary())
```

6. Conclusion

TreeMMM demonstrates that tree-based ensembles with SHAP attribution deliver near-Oracle attribution accuracy on non-linear panel data without requiring the analyst to pre-specify interactions or distributional families. On three non-linear benchmark DGPs, TreeMMM achieves 18.3% non-linear average attribution MAPE — within 1 pp of the Oracle baselines (GLMM-Oracle 17.3%, PyMC-Hier-Oracle 17.5%) that have been given the correct interaction structure and likelihood in advance. In practice, that Oracle knowledge is rarely available; TreeMMM recovers it automatically, discovering 5 of 6 planted channel interactions without manual specification (precision 0.42, recall 0.83, F1 0.56 against 70 total candidate pairs). Against Naive baselines sharing the same underspecification problem, TreeMMM shows a 24% relative improvement (18.3% vs. 24.0%/24.1% MAPE).

A distributional-GLM ablation (Section 3.1) confirms TreeMMM's advantage is driven by interaction discovery and non-linear response learning, not by a distributional-family mismatch: the properly specified GLMMDist-Naive achieves 25.2% — marginally *worse* than GLMM-Naive, further behind TreeMMM.

The mROI module produces budget recommendations validated against ground truth: Spearman $\rho = 0.96$ for channel ranking and 94% direction accuracy. On the pharma benchmark, the optimizer produced +6.1% true lift from equal-budget reallocation, validated against the known DGP.

Our two-Bayesian-baseline design isolates the aggregation cost from the inference paradigm: PyMC-Marketing (70.7% non-linear MAPE) reflects the structural disadvantage of modeling 36 aggregate rows, while PyMC-Hier-

Naive (24.1%) confirms the inference paradigm itself is essentially flat at fixed structure (within 0.5 pp of GLMM-REML on every dataset). A 4× prior-variance sweep finds max attribution-share swings of 0.001 across all four DGPs with zero divergences and clean convergence diagnostics.

TreeMMM is strongest when the analyst wants **discovery over confirmation**: finding patterns they did not hypothesize rather than estimating parameters for patterns they already specified, and when the same pipeline must scale across a portfolio of brands without per-brand manual specification.

Future work (v0.2). Multi-seed evaluation with standard errors across 20+ seeds; Monte Carlo Shapley ground truth; Weibull adstock kernel with joint Optuna decay learning; per-customer random slopes in the Bayesian baseline; real-world validation on anonymized commercial datasets; full-scale power-analysis sweep.

Code availability: `github.com/jamesyoung93/treemmm` (MIT License).

7. Methodology

7.1. Distribution-Aware Objective Selection

TreeMMM supports four objectives matched to outcome distributions (Table 18). An automated diagnostic examines discreteness, zero-inflation rate, skewness, and the mean-variance relationship to recommend an objective; the user can override.

7.2. Link-Function-Aware Attribution Decomposition

SHAP TreeExplainer [8] computes values in margin space:

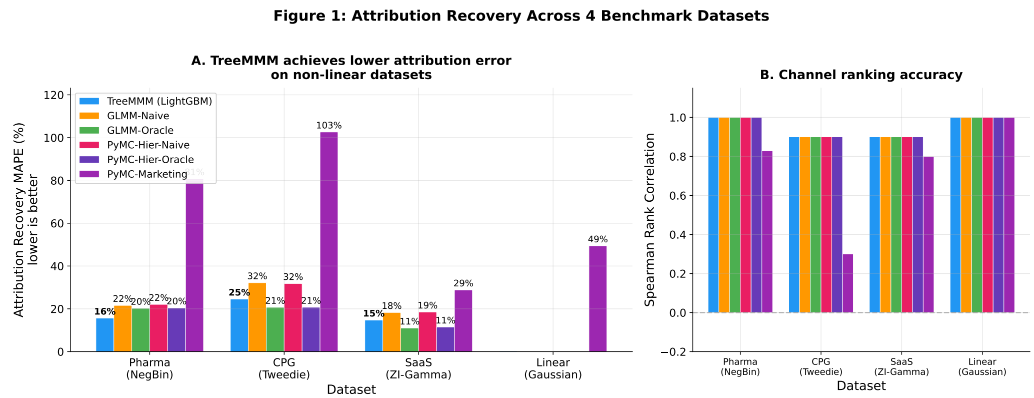


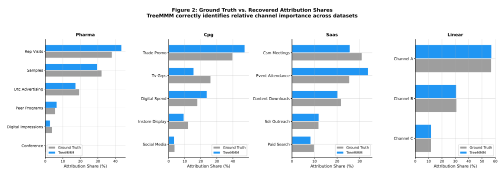
Figure 1: Attribution recovery across models and datasets. Bars show share-MAPE; lower is better. TreeMMM (orange) achieves 17.9% non-linear average MAPE vs. 22.2% for GLMM-Naive (multi-seed, N = 5 seeds). PyMC-Marketing (grey) incurs high error due to aggregation to 36 rows.

- **Identity link** (Gaussian): $\mathbb{E}[y] + \sum_i \phi_i = \hat{y}$. SHAP values are directly additive on the response scale.
- **Log link** (Poisson/Tweedie/Gamma): $\phi_0 + \sum_i \phi_i = \log(\hat{y})$, where ϕ_0 is the mean model prediction in log space. SHAP values are additive on the log scale.

For log-link models, naïve exponentiation breaks additivity: $e^{a+b} \neq e^a + e^b$. TreeMMM's decomposer uses *unsigned proportional allocation*:

$$\alpha_i = \frac{|\phi_i|}{|\phi_0| + \sum_j |\phi_j|} \cdot \hat{y} \tag{2}$$

This guarantees: (1) all attributions are non-negative; (2) attributions sum to $\hat{y}$ for every observation; and (3) the relative channel ranking from log-space SHAP is preserved. A unit test verifies property (2) for all four objectives.



(a) Attribution shares by channel.

(b) Interaction detection precision-recall.

Figure 2: Left: per-channel attribution shares from TreeMMM vs. ground truth on the pharma benchmark. Right: interaction detection precision/recall trade-off at varying SHAP importance thresholds.

7.3. Temporal Cross-Validation

TreeMMM uses time-respecting validation to prevent future data leakage. Two schemes are supported: *rolling origin* (training window grows forward; test window is the next period) and *period-jump-forward* (training window grows; test window jumps by a fixed stride). The minimum training fraction is configurable (default: 50% of periods). No future observation ever appears in the training set.

7.4. Hyperparameter Tuning

Optuna [13] Bayesian optimization tunes tree hyperparameters within each CV fold: number of leaves, learning rate, regularization (L1/L2), feature/bagging fraction, and minimum child samples. The tuning objective is distribution-matched (Poisson deviance for counts, Tweedie deviance for zero-inflated outcomes, MSE for Gaussian).

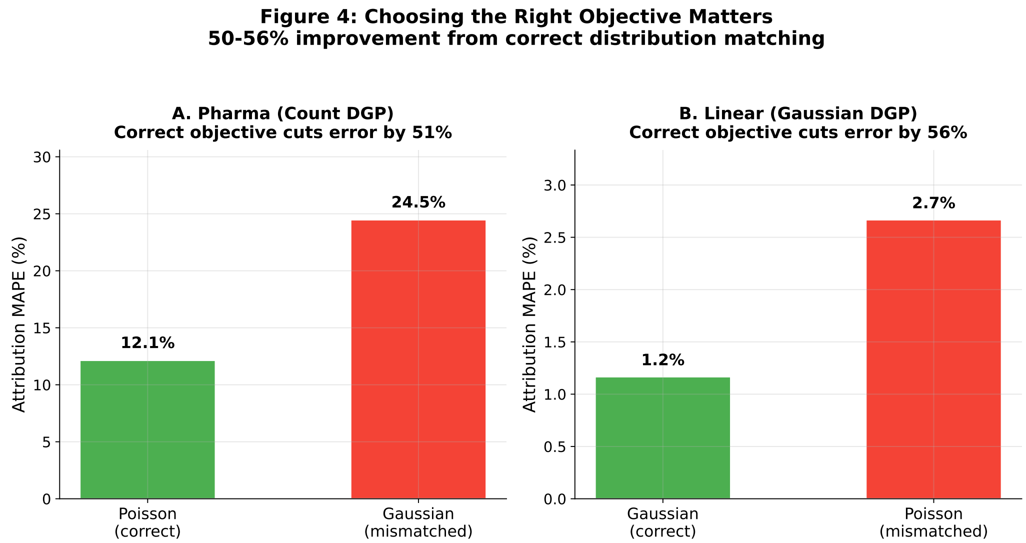


Figure 3: Distribution matching: correct vs. mismatched objective. Using the correct objective (Poisson for count outcomes, Gaussian for symmetric outcomes) improves attribution MAPE by 50–56% relative.

7.5. mROI Simulation with Extrapolation Safety

TreeMMM's marginal Return on Investment (mROI) simulator estimates response curves with a critical safety constraint: per-customer caps are set at the observed 95th percentile, and higher aggregate engagement is achieved by spreading to more customers rather than pushing any individual beyond observed bounds. Every customer-level prediction stays within the training distribution. The constrained optimizer (scipy SLSQP) surfaces customers with untapped marginal sensitivity, revealing that "see your top customers most" is often suboptimal.

7.6. Baseline Models

We compare TreeMMM against five baselines spanning three paradigms:

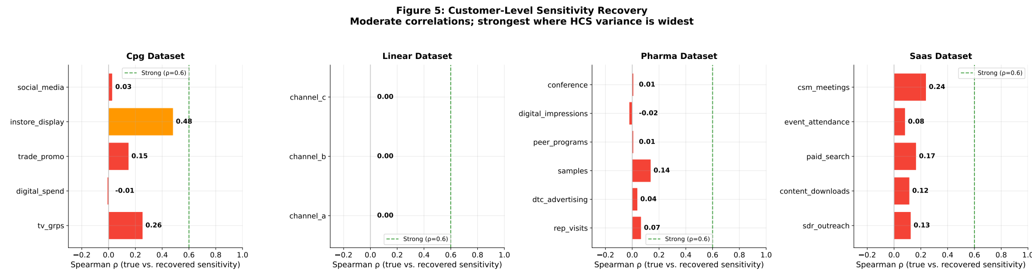


Figure 4: Heterogeneous Customer Sensitivity (HCS) recovery: Spearman ρ between true latent customer sensitivity and customer-level mean $|\phi_i|$. Strongest recovery on CPG in-store display ($\rho = 0.48$).

GLMM-Naive: Main effects only, random intercepts per customer (statsmodels MixedLM). Represents a typical analyst who does not specify interactions or distributional families.

GLMM-Oracle: Correctly specified interaction terms matching the DGP (statsmodels MixedLM). Represents an analyst with perfect interaction knowledge but without distributional family matching.

Both GLMM configurations use statsmodels MixedLM, a Gaussian linear mixed model. For non-Gaussian datasets, the outcome is log-transformed before fitting (`log1p`), approximating a log-normal model. A properly specified distributional GLMM (e.g., `glmmTMB` in R) is not equivalent to this workaround; we discuss this limitation in Section 4.5.1.

PyMC-Hier-Naive: Hierarchical Bayesian linear MMM on the full panel (3,000 entities $\times$ 36 periods, $\sim$108,000 rows), main effects only, customer-level random intercept. Prior structure (after standardization): $\alpha \sim \mathcal{N}(0, 5)$, $a_{\text{cust}} \sim \mathcal{N}(0, \sigma_{\text{cust}})$ with $\sigma_{\text{cust}} \sim \text{HalfNormal}(2)$, $\beta_{\text{main}} \sim \mathcal{N}(0, 1)$, $\sigma_{\text{obs}} \sim \text{HalfNormal}(1)$. Posterior sampling uses `nutpie` (Rust-backed NUTS) with 4 chains, 1,000 tuning + 1,000 draws.

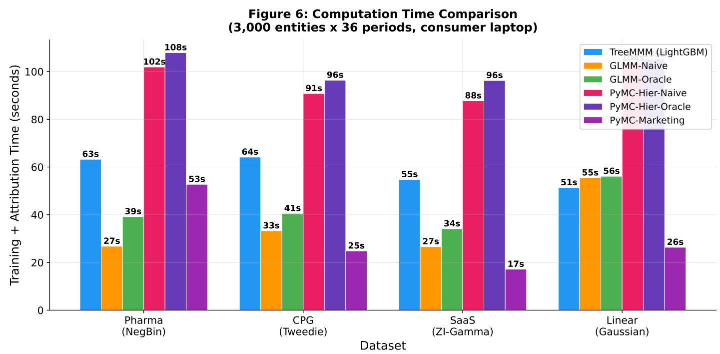


Figure 5: Computation time per dataset at 3,000×36 benchmark scale. TreeMMM: 51–64 s; GLMM variants: 27–56 s; PyMC-Hier: 88–108 s; PyMC-Marketing: 17–53 s (on 36 aggregate rows).

PyMC-Hier-Oracle: Identical to PyMC-Hier-Naive with planted DGP interaction terms added as fixed effects ($\beta_{\text{inter}} \sim \mathcal{N}(0, 0.5)$). The Bayesian counterpart of GLMM-Oracle and the apples-to-apples panel Bayesian competitor to TreeMMM.

PyMC-Marketing [?] (v0.19): The leading open-source Bayesian MMM, implementing NUTS sampling with geometric adstock and logistic saturation transforms. Operates on aggregate time-series (one row per period); our panel data is collapsed to 36 rows. We use default priors, `GeometricAdstock(l_max=4)`, `LogisticSaturation()`, and the NumPyro sampler (500 draws, 500 tuning, 2 chains). Attribution shares are extracted via `compute_channel_contribution_original_scale()`.

The three PyMC variants differ on several key axes summarized in Table 19. A few axes deserve flagging: PyMC-Marketing's built-in adstock and saturation give it parametric carryover and diminishing-returns curves, but it cannot exploit them on 36-row data. PyMC-Hier, conversely, is structurally

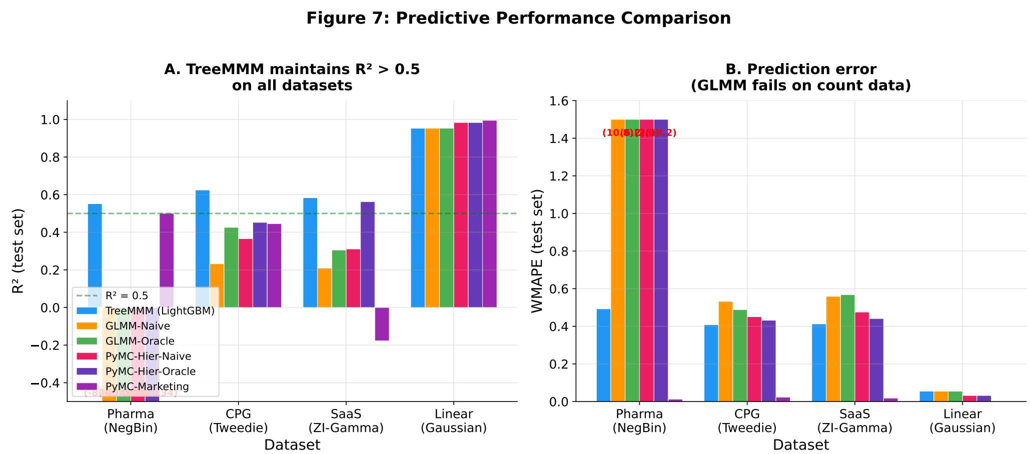


Figure 6: Predictive accuracy (R^2) by model and dataset. TreeMMM achieves $R^2 > 0.5$ on all datasets. GLMM-Naive shows catastrophically negative R^2 on pharma ($-826{,}676$) due to log-transform MixedLM misspecification.

simpler but has 3,000× more data.

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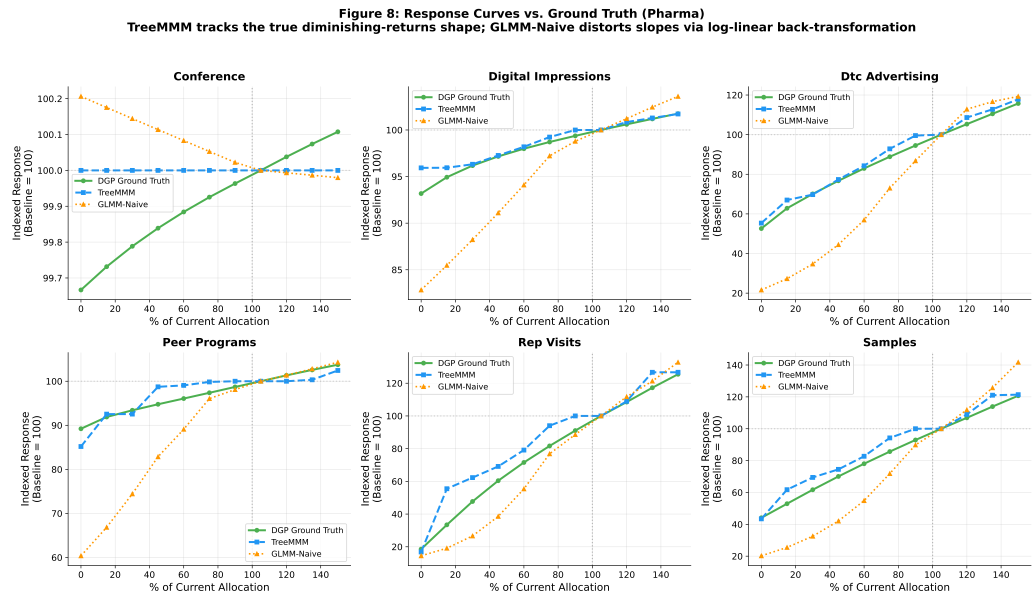

Figure 7: mROI response curves: model-predicted (solid) vs. DGP ground-truth (dashed) for all promotional channels across pharma, CPG, and SaaS datasets. Mean Pearson $r > 0.92$ on all non-linear DGPs.

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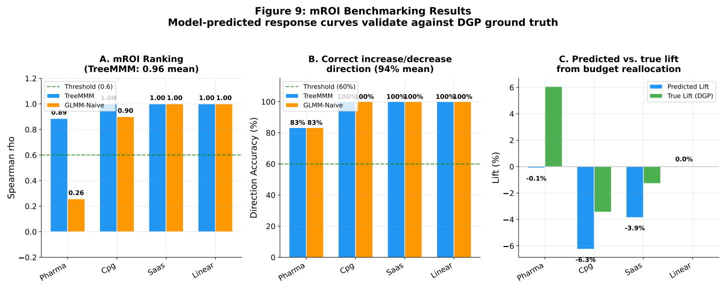

Figure 8: mROI accuracy summary: Spearman ρ between model-predicted and true mROI rankings across all datasets. TreeMMM mean $\rho = 0.96$ on non-linear DGPs.

URL https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

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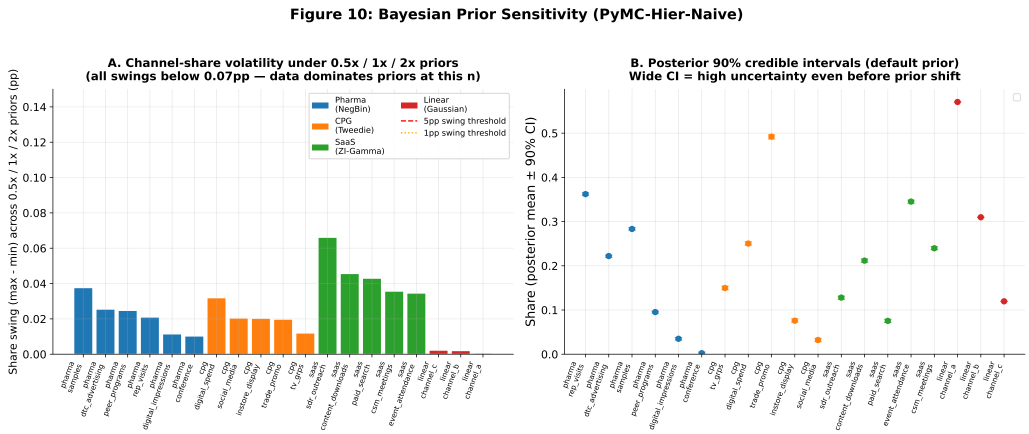

Figure 9: Prior sensitivity sweep: per-channel attribution-share swing across 4× prior-variance bracket (0.5×–2× default sigma, PyMC-Hier-Naive). Max swing across all four DGPs: 0.1 pp (SaaS). The data dominates the prior at this sample size.

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URL https://arxiv.org/abs/1911.11718

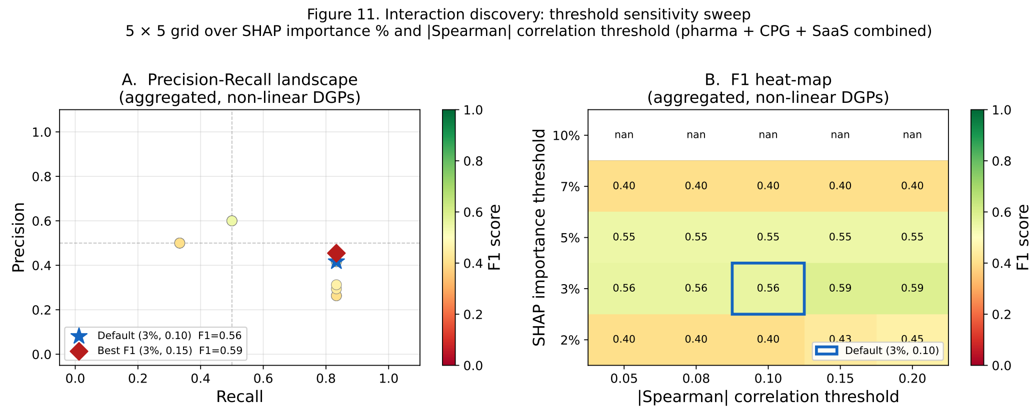

Figure 10: Interaction detection threshold sensitivity: F1/Precision/Recall over a 5×5 grid of SHAP importance thresholds (2–10%) and Spearman $|\rho_s|$ thresholds (0.05–0.20). F1 is stable (plateau from 2–5% importance); recall drops sharply above 7%.

[13] T. Akiba, S. Sano, T. Yanase, T. Ohta, M. Koyama, Optuna: A next-generation hyperparameter optimization framework, arXiv preprint arXiv:1907.10902 (2019).

Appendix A. Exploratory Neural MMM Comparison (DeepCausalMMM)

We include an exploratory comparison with DeepCausalMMM [3], a neural MMM combining GRU temporal encoding, learned DAG structure, Hill saturation curves, and time-varying coefficients. This appendix is presented separately because structural limitations in the evaluation design make it less directly comparable to the regression baselines in the main text.

Appendix A.1. Configuration and Data Reshaping

DeepCausalMMM was designed for DMA-level aggregated data (∼190 geographic regions × 109 weeks). Our benchmark uses customer-level panel data (3,000 entities × 36 periods). To create a comparison, we reshaped the panel data into 3D tensors [regions, timesteps, channels] by treating each customer as a "region." Training used DeepCausalMMM's `UnifiedDataPipeline` and `ModelTrainer` with reduced hyperparameters (Table A.20).

Appendix A.2. Fairness Caveats

This comparison favors TreeMMM in several structural ways:

1. **Data format mismatch**: customers-as-regions violates DeepCausalMMM's geographic aggregation design.
2. **Hyperparameter downgrading**: five key hyperparameters were reduced; no Optuna tuning was performed for DeepCausalMMM.
3. **Subsampling**: only 500 of 3,000 customers were used.
4. **No distribution awareness**: DeepCausalMMM uses Huber loss (not matched to count or zero-inflated outcomes).

A fair comparison would require (a) geographic-level time-series aggregation, (b) default hyperparameters with full tuning, and (c) longer time series (100+ periods).

Appendix A.3. Results

DeepCausalMMM shows high MAPE on count-valued (pharma: 91.5%) and Tweedie (CPG: 112.5%) outcomes, suggesting its Huber loss is poorly matched to these distributions. On SaaS (continuous-like), it performs more competitively (21.9% vs. TreeMMM's 14.7%). On the linear DGP, DeepCausalMMM introduces spurious non-linearity (62.4% MAPE), suggesting optimization difficulties. Notably, rank correlation on pharma improved at full scale (0.89 at 3,000 $\times$ 36 vs. 0.49 at 100 $\times$ 12), suggesting the architecture can learn relative channel importance even when absolute shares are poorly calibrated. This is, to our knowledge, the first evaluation of any neural MMM against known attribution ground truth.

Table 14: Adstock-planted headline DGPs — Attribution MAPE (%) by model and preprocessing mode. Lower is better. Bold = best MAPE per dataset. N = 200×18, seed = 42. Results from `paper/results/benchmark_adstock_headline.csv`.

Dataset	Model	No-adstock MAPE	Adstock-aware MAPE	Δ (pp)
pharma	**TreeMMM**	**15.3**	42.4	−27.1
pharma	GLMM-Naive	44.1	46.2	−2.1
pharma	GLMM-Oracle	46.6	55.7	−9.1
pharma	PyMC-Hier-Naive	38.8	57.1	−18.3
pharma	PyMC-Hier-Oracle	44.6	60.3	−15.7
pharma	PyMC-Marketing	286.6	313.4	−26.8
cpg	TreeMMM	56.6	72.3	−15.7
cpg	**GLMM-Naive**	62.8	**50.1**	+12.7
cpg	GLMM-Oracle	73.8	65.1	+8.7
cpg	**PyMC-Hier-Naive**	62.0	**34.8**	+27.2
cpg	**PyMC-Hier-Oracle**	72.0	**46.9**	+25.1
cpg	PyMC-Marketing	247.2	309.7	−62.5
saas	**TreeMMM**	**52.2**	110.4	−58.2
saas	GLMM-Naive	55.0	182.1	−127.1
saas	GLMM-Oracle	75.0	109.5	−34.5
saas	PyMC-Hier-Naive	51.1	173.3	−122.2
saas	PyMC-Hier-Oracle	75.3	118.1	−42.8
saas	PyMC-Marketing	221.9	227.9	−6.0

Δ = No-adstock MAPE − Adstock-aware MAPE; negative = preprocessing hurts.

Table 15: Geo-panel DGP channel specifications.

Channel	Adstock decay	Saturation form	Effect weight
`tv_grps`	0.50	Logistic (k=0.04, x_0=50 GRPs)	1.8
`digital_spend`	0.30	Logistic (k=0.08, x_0=\$20k)	1.4
`trade_promo`	0.00	Linear (no saturation)	0.6

Table 16: Geo-panel comparison: Attribution MAPE (%), Rank ρ, R^2, and WMAPE by model. Results from `paper/results/benchmark_geo_panel.csv`, run 2026-05-10.

Model	Attr. MAPE	Rank ρ	R^2	WMAPE	Fit (s)	Notes
TreeMMM-Adstock	**29.7**	**1.00**	0.08	0.35	25	Planted decays us
PyMC-Marketing (weak priors)	52.1	0.50	−0.53	0.08	115	66 divergences
PyMC-Marketing (informative)	52.1	0.50	−0.52	0.08	61	Prior API limited
GLMM-Aggregate	215.0	−1.00	0.58	0.04	1	Coefficient rank r
Robyn	n/a	n/a	n/a	n/a	n/a	R-only package
Meridian‡	57.0	1.00	—	—	1,734	EagerTensor; no

‡ Meridian MCMC completed (2 chains, 500 keep, ∼29 min). Attribution shares via `Analyzer.incremental_outcome().numpy()`: tv = 70.5% (true 44.9%), digital = 27.1% (true 37.5%), trade = 2.4% (true 17.5%). Rank correct (tv > digital > trade). R^2/WMAPE not extractable.

Table 17: Attribution MAPE (%) by model, dataset, and scale. Lower is better. Run `python paper/run_power_analysis.py` to populate dashes (full sweep pending).

Dataset	Model	n=200	n=500	n=1500	n=3000
Pharma (NegBin)	TreeMMM (LightGBM)	—	—	—	14.5
	GLMM-Naive	—	—	—	19.2
	GLMM-Oracle	—	—	—	25.2
	PyMC-Hier-Naive	—	—	—	20.6
CPG (Tweedie)	TreeMMM (LightGBM)	—	—	—	22.5
	GLMM-Naive	—	—	—	29.0
	GLMM-Oracle	—	—	—	21.9
	PyMC-Hier-Naive	—	—	—	29.0
SaaS (ZI-Gamma)	TreeMMM (LightGBM)	—	—	—	16.7
	GLMM-Naive	—	—	—	18.4
	GLMM-Oracle	—	—	—	12.0
	PyMC-Hier-Naive	—	—	—	18.5
Linear (Gaussian)	TreeMMM (LightGBM)	—	—	—	0.4
	GLMM-Naive	—	—	—	0.3
	GLMM-Oracle	—	—	—	0.3
	PyMC-Hier-Naive	—	—	—	0.1

n=3000 column is from the full 5-seed multi-seed benchmark (Table 4); other columns pending full sweep. Periods: 12 (n=200), 24 (n=500), 36 (n=1500, 3000).

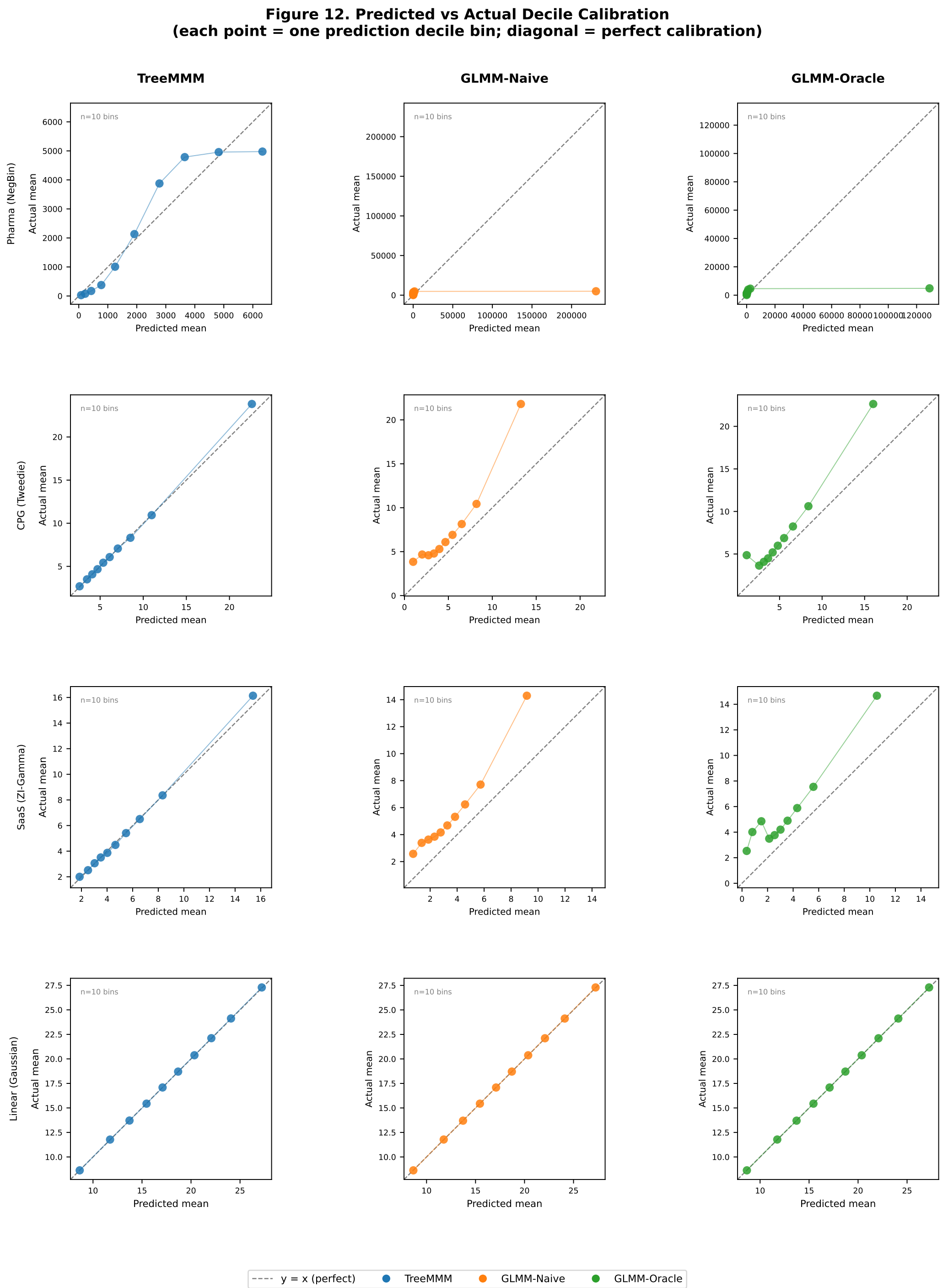

Figure 11: Predicted-vs-actual decile calibration plots for TreeMMM, GLMM-Naive, and GLMM-Oracle across all four DGPs. GLMM-Naive's pharma MAD = 23,977 (47× overestimate in top decile); TreeMMM pharma MAD = 503 (47× tighter). All three models are well-calibrated on the linear DGP.

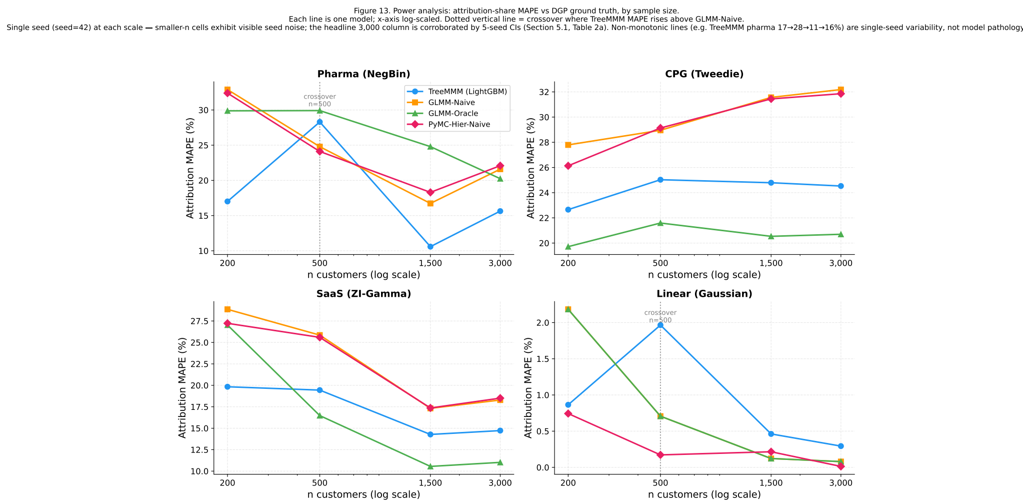

Figure 12: Power analysis: Attribution MAPE vs. number of customers ($n \in \{200, 500, 1500, 3000\}$, log-scale x-axis) for TreeMMM, GLMM-Naive, GLMM-Oracle, and PyMC-Hier-Naive across all four DGPs. Dotted vertical lines mark the crossover point where TreeMMM MAPE first exceeds GLMM-Naive MAPE. (Full sweep pending; n=3000 column populated.)

Table 18: Outcome distributions and corresponding TreeMMM objectives.

Distribution	Objective	When to Use
Gaussian	MSE	Continuous, symmetric (revenue, value sales)
Poisson	Log-link	Non-negative counts (Rx, orders, NPS)
Tweedie	Log-link	Zero-inflated continuous (revenue with stockouts)
Gamma	Log-link	Strictly positive continuous (per-transaction revenue)

Table 19: PyMC variants used in the benchmark and their key differences.

Axis	PyMC-Hier-Naive	PyMC-Hier-Oracle	PyMC-Marketing
Data shape	Panel (108K rows)	Panel (108K rows)	Aggregate (36 rows)
Random effects	Per-customer intercept	Per-customer intercept	None
Fixed effects	Promo + controls	+ planted interactions	Promo + controls
Adstock	None	None	`GeomAdstock(l_max=4)`
Saturation	None	None	`LogisticSat()`
Likelihood	$\mathcal{N}$ on `log1p(y)`	Same	$\mathcal{N}$ on raw y
Sampler	`nutpie` NUTS	`nutpie` NUTS	NumPyro NUTS
Draws/tune/chains	1000/1000/4	1000/1000/4	500/500/2
Wall-clock (3K×36)	90–105 s	90–105 s	25–55 s
Non-linear MAPE	24.1%	17.5%	70.7%

Table A.20: DeepCausalMMM configuration vs. published defaults.

Parameter	Default	Our Config	Reason
Epochs	1,500	800	Benchmark speed
Hidden dim	280	128	Fewer channels
Patience	300	200	Proportional reduction
Regions	190 DMAs	500 (subsampled)	Panel has 3,000 entities
Time periods	109 weeks	36 months	Benchmark design

Table A.21: Exploratory neural MMM comparison (500-customer subsample; TreeMMM MAPE differs from Table 5 due to subsampling).

Dataset	TreeMMM MAPE	DeepCausalMMM MAPE	DCausal R^2	DCausal Rank
Pharma (NegBin)	16.4%	91.5%	0.12	0.
CPG (Tweedie)	25.0%	112.5%	−0.02	0.
SaaS (ZI-Gamma)	14.7%	21.9%	−0.05	0.
Linear (Gaussian)	0.3%	62.4%	−0.07	−0.
Non-linear avg	**18.7%**	75.3%	0.02	0.